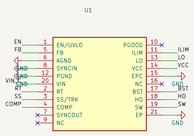
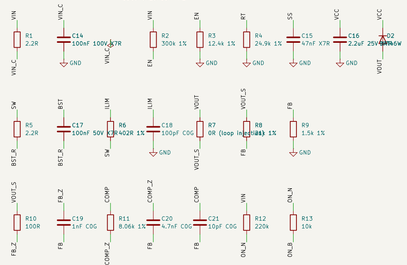
12 V buck: LM5146, 48 V to 12 V, 12 A, 400 kHz





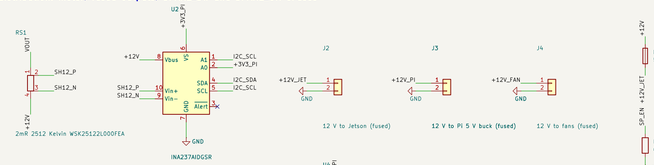


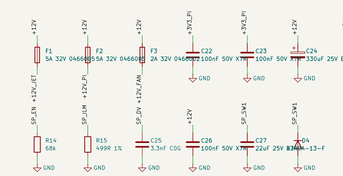


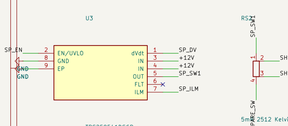


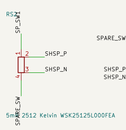
Input: SK810L crowbar (a reversed pigtail clamps at about 1 V; U7 on the main board limits and times out). 1 uF at the connector, 2.2 uH filter, 47 uF aluminium (0.42 Ohm ESR damps the filter and the pigtail). 5 x 2.2 uF hot-loop ceramics at Q1. tools/absorber_sim.py (main board) runs the bank-trip spike through U7, F2, the pigtail and this filter.
UVLO: 300k / 12.4k: on at 30.2 V, off at 27.2 V (10 uA hysteresis current). Below the 30.5 V worst sag at 90 A from a 38 V pack, so motor load can't brown out compute before the BMS does; the Pi still shuts down on its load-aware threshold.
ON/OFF: the main board's 2N7002 pulls pin 5 low through 220R when U7's PGD is good. Pin 5 open: 220k from VIN biases the NPN, which holds EN low (off). Pin 5 low: NPN off, EN rises; on. Pin 5 idles at about 2.7 V, like the DVE.
Power rise (Ti design 2 upgrade): 400 kHz (RT 24.9k). 6.8 uH 8.9 mOhm 21.8 A isat. 3.3 A ripple: Q1 BSC070N10NS5 (low Qgd). Q2 BSC040N10NS5. Valley limit by Q2 RDS(on): 200 uA x 4028 / 4.5 mOhm = 18 A cold, about 12–13 A hot: over the 11.6 A ceiling, under the 19–21 A of the inductor. Hiccup on overload.
Output: 12.0 V (0.8 V x (1 + 21k / 1.5k)). 5 x 22 uF at VOUT plus the 330 uF bulk cap on +12V behind RS1 (12 V distribution, below). Compensation is Ti's for this L and 5 x 22 uF only (40 kHz crossover): recheck it for the 330 uF and measure the loop with an injection resistor in place of the 0R before loading. VCC from VOUT through BAT46W (Ti's D1) cuts bias loss.
Loss at 12 A, 48 V (est): Q1 0.4 W conduction + 1.2 W switching, Q2 0.7 W, inductor 1.3 W, controller 0.3 W: about 4 W, 97 %. Board: about 60 x 85 mm with the 12 V distribution and its six outputs, bolted to the rail with a thermal pad like the DVE.

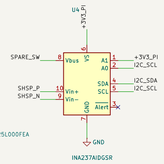
12 V distribution: meter, fused outputs, SPARE_SW and SPARE_ON eFuses

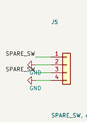


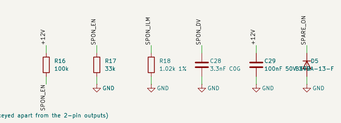


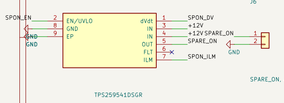














Moved from the main board (split S1): the 12 V meter (0x40) and SPARE_SW meter (0x47) on the Pi's I2C, powered by the Pi's 3.3 V through J1 pin 6; pull-ups only at the Pi.
SP_EN (J1 pin 9, between GND pins) comes from the main board's RUN through 100k and a series diode; the 68k here pulls EN low, so SPARE_SW is off when RUN drops, the diode opens or the pigtail is unplugged (fail-off). EN about 2.6 V with RUN up.
SPARE_SW: limit 3.87–4.42 A (499R), dV/dt 3.3 nF about 12 V/ms. SPARE_ON: unswitched, about 2 A limit. B340A clamps on both outputs. SMBJ15A: servo back-feed clamp on +12V.
330 uF + 5 x 22 uF now sit at the converter output, not at the far end of the pigtail: recheck the compensation and measure the loop at the 0R injection point before loading.
The DVE fallback no longer plugs into J24: this board has no 12 V input, and the old DVE adapter pinout puts 12 V on SCL and SP_EN.

On the J24 pigtail (split S1 pinout). Main board: hardware/power-board/schematic.

marengo	
Sheet: /	
File: marengo_buck12.kicad_sch	
Title: marengo 12 V buck board (rev B5)	
Size: A1 Date: 2026-11-06	Rev: B-wip
KiCad E.D.A. 10.0.8	Id: 1/1